MINOR PROJECT REPORT

Student Performance Analysis and Prediction System

A Python-based data analysis and machine learning project

Student Name	Sujal Ghimire
Roll Number	AIML-A6/SEP-11832
Program	AI/ML with Python Training Program
Organization	InternsElite
Project Type	Minor Project

Submitted as part of the internship training project requirements.

Abstract

This project analyzes student academic records and builds a Linear Regression model to estimate final examination marks. The workflow includes dataset inspection, cleaning, feature selection, exploratory data analysis (EDA), model training, evaluation, cross-validation, and a sample prediction. The selected inputs are weekly study-time category, number of absences, first-period marks (G1), and second-period marks (G2). The target is final-period marks (G3). The saved model and visualizations form part of the project deliverables.

1. Introduction

Academic performance can be associated with several factors, including study habits, attendance-related records, and previous examination performance. This project uses historical student data to explore these relationships and demonstrate how a supervised machine learning model can predict a numerical academic outcome. The model is intended as an analytical aid, not as a definitive judgement of a student's ability.

2. Problem Statement

Develop a Python-based machine learning system that analyzes student academic information, explores patterns in the data, and predicts final examination marks from selected academic features.

3. Objectives

- Load and inspect a student performance dataset using Pandas and NumPy.
- Check for missing values and duplicate records and prepare a cleaned dataset.
- Calculate descriptive statistics and create the required visualizations.
- Train a Linear Regression model using selected input features.
- Evaluate model performance using MAE, RMSE, R^2, and 5-fold cross-validation.
- Generate a sample prediction and save the trained model.

4. Dataset Description

The submitted CSV contains 395 records and 33 columns. It is the Mathematics student performance dataset, with academic and demographic fields. This implementation uses four predictors: studytime, absences, G1, and G2. The target variable is G3. G1, G2, and G3 are examination grades on a 0–20 scale. The studytime field is an ordinal category (1–4), not a direct number of hours. The absences field records absence counts; it is used as an attendance-related proxy because the dataset does not provide an attendance percentage.

Dataset source: UCI Machine Learning Repository, Student Performance dataset (ID 320). The CSV used in the notebook was loaded from a hosted copy of the UCI data. Dataset information: https://archive.ics.uci.edu/dataset/320/student+performance

5. Tools and Technologies

Python; NumPy; Pandas; Matplotlib; Seaborn; Scikit-learn; Joblib; Google Colab/Jupyter Notebook.

6. Data Preprocessing and Feature Selection

The dataset was inspected for missing values and duplicate rows. The submitted dataset contains 0 missing values and 0 duplicate rows. A cleaned copy was created using duplicate removal and missing-row removal; because the checks returned zero for both, the cleaned dataset retained all 395 records. The feature matrix X contains studytime, absences, G1, and G2. The target vector y contains G3.

7. Exploratory Data Analysis

Descriptive statistics (mean, median, and standard deviation) were calculated for studytime, absences, G1, G2, and G3. The following charts were generated and saved: average examination marks bar chart, final-mark histogram,

study-time versus final-mark scatter plot, final-mark box plot, and numerical-feature correlation heatmap.

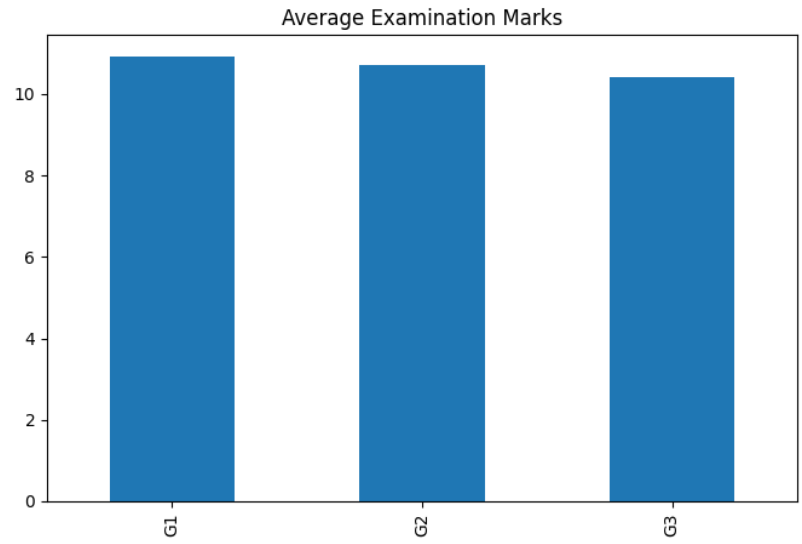


Figure 1. Average marks in G1, G2, and G3.

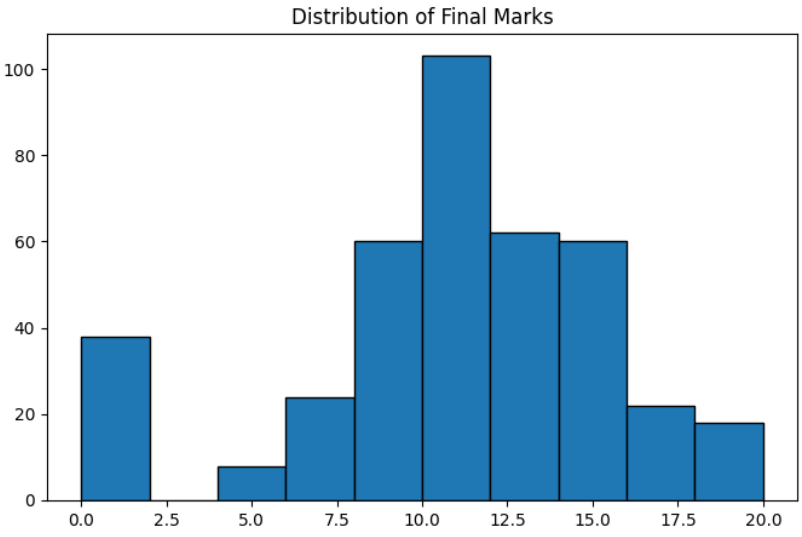


Figure 2. Distribution of final examination marks (G3).

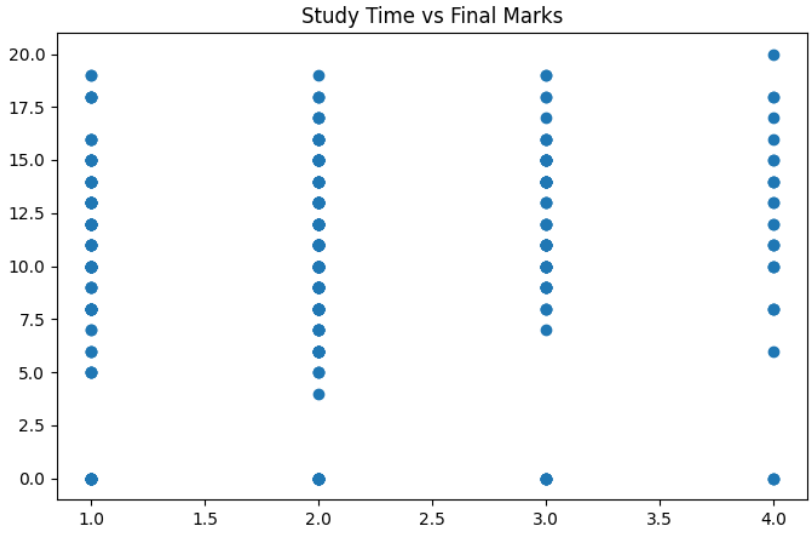


Figure 3. Study-time category versus final examination marks.

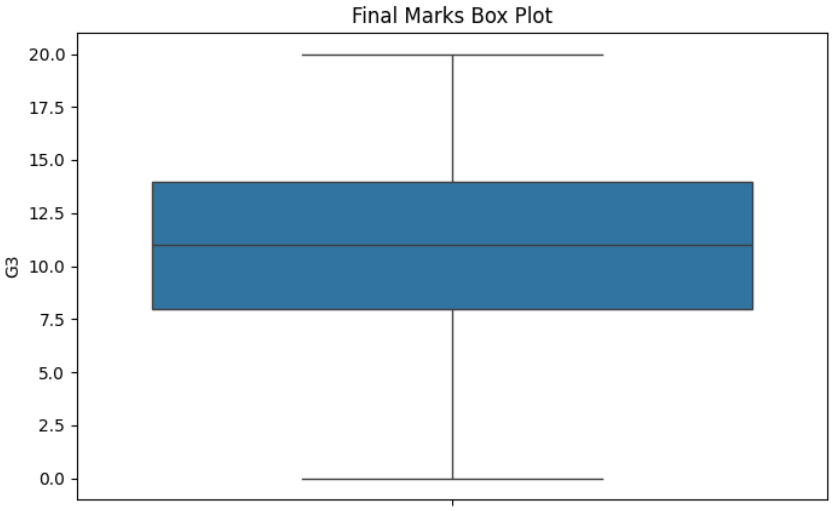


Figure 4. Box plot of final examination marks.

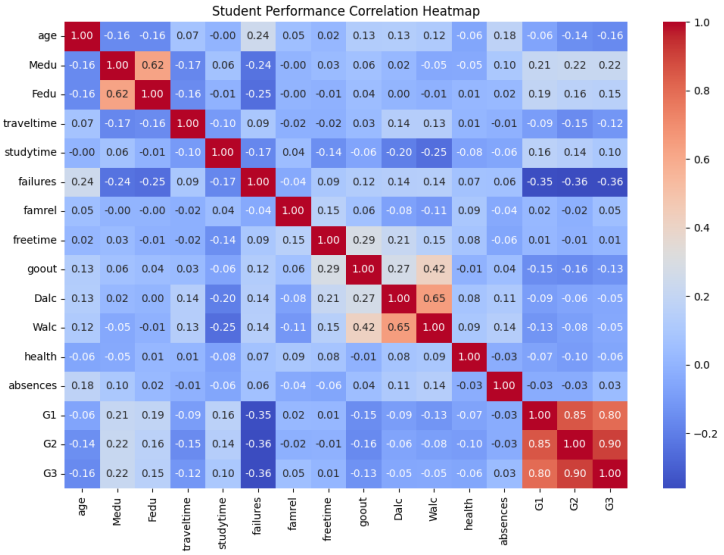


Figure 5. Correlation heatmap for numerical variables.

8. Machine Learning Methodology

A Linear Regression model was selected because the target variable (G3) is numerical. The dataset was split into training and testing subsets using an 80:20 split and random_state=42. This produced 316 training records and 79 testing records. The model was fitted on the training subset and predictions were generated for the held-out test subset.

9. Model Evaluation

Metric	Observed result	Interpretation
Mean Absolute Error (MAE)	1.3126	Average absolute prediction error is about 1.31 marks.
Root Mean Squared Error (RMSE)	1.8143	Larger errors receive greater weight.
R² score (test set)	0.7962	The model explains about 79.62% of test-set target variation.
Mean 5-fold CV R²	0.8292	Average R² across five folds of the training data.

The five cross-validation R² scores recorded during execution were 0.8253, 0.8564, 0.8666, 0.8064, and 0.7911, with an average of 0.8292. The held-out test score and cross-validation average are separate estimates and should not be treated as interchangeable.

10. Sample Prediction

A sample student record was entered with studytime=3, absences=4, G1=12, and G2=14. The trained model returned a predicted G3 score of 13.71 out of 20. This is a model estimate, not an actual examination result. The displayed prediction was clipped to the valid 0–20 grade range.

11. Model Persistence

The trained Linear Regression model was saved using Joblib as student_performance_model.pkl. The saved artifact was then loaded and used to generate a test prediction, demonstrating that the model can be reused without retraining in the same workflow.

12. Limitations

• The dataset represents a particular student population and subject context; results may not generalize to other schools, regions, or subjects.
• Studytime is an ordinal category, not an exact number of study hours.
• Absences are not the same as attendance percentage.
• The model uses G1 and G2, which are strongly related to the final grade; predictions made before those assessments would require a different feature set.
• Linear Regression models linear relationships and may not capture all factors affecting student performance.
• The model identifies statistical patterns and should not be used as the sole basis for decisions about students.

13. Future Improvements

Future work could compare additional regression algorithms, tune model settings using training data only, test on an independent dataset, add attendance percentage where reliable data is available, and build a simple user interface for entering student details and viewing predictions.

14. Conclusion

The project completed a full introductory machine learning workflow for student performance analysis. The dataset was inspected and cleaned, five EDA visualizations were produced, and a Linear Regression model was trained and evaluated. On the held-out test set, the model achieved MAE 1.3126, RMSE 2.0443, and R^2 0.7962. Five-fold cross-validation on the training set produced an average R^2 of 0.8292. The sample prediction was 13.71/20. These results demonstrate the practical use of Python data analysis and supervised learning while highlighting the importance of interpreting model limitations.

15. References

1. UCI Machine Learning Repository. Student Performance Dataset. https://archive.ics.uci.edu/dataset/320/student+performance
2. Scikit-learn documentation. LinearRegression, train_test_split, and model evaluation metrics. https://scikit-learn.org/stable/
3. Pandas documentation. https://pandas.pydata.org/docs/
4. Matplotlib documentation. https://matplotlib.org/stable/
5. Seaborn documentation. https://seaborn.pydata.org/

Appendix: Project artifacts

The accompanying project files include the cleaned dataset CSV, the saved Joblib model artifact, and a ZIP archive containing the five EDA visualizations. The notebook/source code should also be included in the public project repository as required by the internship submission guidelines.